

Unit II — Combinatorics

Counting Symptom Sets: Exactly, At Least, and Ordered Selections

GROUP 4 • Discrete Mathematics • August 28, 2026

Educational mathematics assignment — not medical advice.

1. Case Title (Unit II Focus)

Sizing the search space when the triage engine narrows a 12-symptom database to a 5-symptom core.

2. Problem Statement (Unit II)

For $n = 12$, $k = 5$: how many distinct sets can the system examine under three policies — exactly five, at least five, and ordered five? Under-counting wastes coverage; over-counting explodes storage/UI.

3. Real-world Context

A 12-item checklist is filtered to five for a rapid screen. Capacity planners must know how many test cases to generate and whether a ranked selector (order matters) is worth its 120× cost.

4. Mathematical Concept Used

Binomial coefficients $C(n,k)=n!/(k!(n-k)!)$, permutations $P(n,k)=n!/(n-k)!$, power-set identity $2^n=\sum C(n,k)$, complement principle, and the identity $P(n,k)=C(n,k)\cdot k!$.

5. Mathematical Formulation

Universe U , $|U|=12$. Exactly- k : $C(12,5)$. At-least- k : $\sum_{i=5}^{12} C(12,i)=2^{12}-\sum_{i=0}^4 C(12,i)$. Ordered- k : $P(12,5)$.

6. Analysis / Solution

$C(12,5)=12!/(5!7!)=(12\cdot11\cdot10\cdot9\cdot8)/(120)=95,040/120=792$.

At least five: total 4096 – (1+12+66+220+495)=4096–794=3302. Direct sum: 792+924+792+495+220+66+12+1=3302 ✓.

Ordered five: $P(12,5)=12!7!=12\cdot11\cdot10\cdot9\cdot8=95,040=792\cdot120$. Ordering multiplies by 5!; relaxing “exactly five” to “at least five” multiplies by ~4.17.

7. Diagram / Tables

Three policies as cards and a summary table:

Policy	Formula	Count	Interpretation
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Exactly 5	$C(12,5)$	792	One 5-set, order ignored
At least 5	$\sum C(12,k), k=5\dots 12$	3,302	Any superset of 5
Ordered 5	$P(12,5)$	95,040	5 ranked by severity

8. Interpretation of Results

Middle number 3,302 bounds curation effort; 792 is the UI choice count for a fixed-5 screen; 95,040 warns that ranking needs virtualized rendering.

9. Practical Application

Deployed as the Combination Calculator panel: n/k sliders defaulting to 12/5 display the three counts live via exact integer arithmetic (no floats). Admins explore capacity trade-offs interactively.

10. Innovation / Proposed Improvement

- Scale to $n \approx 50$ with pruning and inclusion-exclusion for at-least-k queries.
- Weighted counting with symptom-frequency priors.

11. Conclusion

Exact counting makes the system auditable and the UI honest about its search space.

12. References

- Rosen, Chap. 6 — Counting.
- Vite + Vitest docs — exact integer arithmetic.

All counts and proofs in this report are computationally verified (see live site <https://ai-medical-diagnosis-site.vercel.app>).